

The de Sitter–Unruh Modified-Inertia Theory: A One-Parameter Account of the MOND Acceleration Scale

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Abstract

I present the de Sitter–Unruh modified-inertia (MI) theory as a self-contained, **one-parameter** physical account of the MOND acceleration scale a_0 — explicitly **not** a theory of everything. The theory rests on two physical inputs. First, an accelerated body immersed in a de Sitter universe sees a thermal bath at the Deser–Levin temperature $T(a) = (\hbar/2\pi k_{\text{Bc}})\sqrt{(a^2 + (cH_{\Lambda})^2)}$, which never falls below a universal floor $T_0 = \hbar c H_{\Lambda}/(2\pi k_{\text{Bc}}) = 2.20 \times 10^{-30}$ K set by the dark-energy horizon alone. Second, a relational/Machian principle: a body’s force-responsive (dynamical) inertia is the **excess** of its bath response above that universal floor — a body unaccelerated relative to the cosmic vacuum carries no dynamical inertia. From these two inputs, four structures follow as consequences rather than fits: (i) the **scale** $a_0 = cH_{\Lambda}/Z \sim \sqrt{\Lambda}$; (ii) the **interpolation form** $\mu_{\text{fw}}(x) = (\sqrt{(1+4x^2)}-1)/2x$, with the strikingly clean inverse $\mu/(1-\mu^2) = a/a_0$, realized as the minimum of a convex, ghost-free potential; (iii) the **memory kernel** $\theta(0) = \sqrt{2}$ (amplitude-branch selection) with a Lorentzian shape forced by the de Sitter Wightman function; and (iv) the **MOND sign** itself — the relational principle selects inertia-weakening (MOND) and rejects the passive-bath, inertia-raising (anti-MOND) reading. What remains is a single free number Z , provably free (κ -closure), entering exactly as Newton’s G enters general relativity. I keep the honesty explicit and two-sided throughout: the sign is **conditional on the Machian premise** (the honest crux); Z and hence a_0 ’s numerical value are **not derived**; and the Standard-Model mass sector stays **walled** — this is a one-parameter theory of one infrared scale, **not a TOE**. The theory is consistent with, and assembles, the previously published effective-field-theory results (DOIs 10.5281/zenodo.20935948, 20938891, 20947913, 20963226) and with the author’s 2026 retraction of all earlier TOE/Standard-Model overclaims. Predictions: an s^{TX} Lorentz-violation dipole, an $a_0(z)$ “hostage” tied to the dark-energy equation of state, a modified-gravity-impossible dwarf orbital-history dispersion signature, and a definite floor temperature $T_0 = 2.20 \times 10^{-30}$ K.

1. The two physical inputs

This paper takes one physical proposal as its subject and reasons forward from that proposal’s own premises. It is not a variant of Milgrom’s MOND and is not to be read through MOND’s lens; where the standard MOND apparatus appears, it appears as a *consequence*, downstream of the two inputs stated here, never as a baseline against which the theory is judged.

1.1 Input A — the de Sitter–Unruh bath and its universal floor

An observer with proper acceleration a in a de Sitter universe sees a thermal bath. Combining the Unruh effect with the Gibbons–Hawking temperature of the cosmic horizon, Deser & Levin (1997) give the temperature such an observer measures as

$$T(a) = (\hbar / 2\pi k_B c) \cdot \sqrt{a^2 + (cH_\Lambda)^2},$$

where H_Λ is the de Sitter rate fixed by the dark-energy density alone — the pure- Λ horizon clock, **not** the instantaneous Hubble rate $H(z)$. Numerically $cH_\Lambda = 5.42 \times 10^{-10} \text{ m/s}^2$ and $H_\Lambda = 1.81 \times 10^{-18} \text{ s}^{-1}$ (a horizon time $1/H_\Lambda \approx 17.5 \text{ Gyr}$).

The structure that does the work is the additive $(cH_\Lambda)^2$ under the root. Even a body at perfect rest ($a = 0$) does not see zero temperature; it sees a universal floor

$$T_0 \equiv T(0) = \hbar c H_\Lambda / (2\pi k_B c) = 2.20 \times 10^{-30} \text{ K}.$$

This floor is the Gibbons–Hawking temperature of the cosmic horizon. It is shared identically by every body and by the geodesic vacuum itself: nothing is colder, or more at rest, than the cosmic vacuum. The floor is universal in the precise sense that $dT_0/da = 0$ — it is the same constant for all bodies and carries no information about any particular body’s state of motion.

This input is established physics in its kinematic content: the Deser–Levin temperature is a real property of de Sitter space. The further *reading* — that this temperature is what a body’s inertia responds to — is the theory’s postulate, made explicit in Input B and again in §5.

1.2 Input B — relational/Machian inertia: dynamical inertia is the excess above the floor

The second input is a principle about what inertia *is*. In a relational (Machian) view, inertia is a body’s resistance to acceleration **relative to the rest of the universe**, not relative to absolute space. The universal, body-independent contribution to the bath response is the shared floor T_0 . The theory’s principle is then sharp:

The force-responsive (dynamical) inertia of a body is the EXCESS of its bath response above the universal de Sitter floor T_0 . A body unaccelerated relative to the cosmic vacuum has zero dynamical inertia.

The kinematically meaningful, body-specific quantity is therefore not $T(a)$ but the floor-subtracted excess. A short computation (sympy-verified) gives the clean fact that drives everything downstream:

$$\sqrt{T(a)^2 - T_0^2} = (\hbar / 2\pi k_B c) \cdot a, \text{ exactly linear in } a.$$

The excess responds to force: $d\sqrt{T^2 - T_0^2}/da|_0 = \hbar/2\pi k_B c$, finite and nonzero. The full temperature does **not** respond at the floor: $dT/da|_0 = 0$. So the force-responsive piece of the inertia is *necessarily* the excess, and necessarily *linear* in the acceleration amplitude — a point that will matter for both the form (§3) and the kernel (§4).

These two inputs — the bath with its floor (A), and the relational excess principle (B) — are the entire physical foundation. Everything in §§2–5 is a consequence of them, together with one shape posit (the $\sqrt{}$ -equation-of-state, stated where it is used) and one free number (§6).

2. The derived scale: $a_0 = cH_\Lambda/Z \sim \sqrt{\Lambda}$

The acceleration at which the bath response becomes order-unity — where the excess $\sqrt{(T^2 - T_0^2)}$ becomes comparable to the floor T_0 itself — is set by the single rate in the problem, cH_Λ . The crossover is

$$a_0 = cH_\Lambda / Z = 9.36 \times 10^{-11} \text{ m/s}^2,$$

with Z the theory's geometric normalization (the lone free number, §6). Because $cH_\Lambda \propto H_\Lambda \propto \sqrt{(\rho_\Lambda)} \propto \sqrt{\Lambda}$, the scale is

$$a_0 = (c/2) \cdot \sqrt{(G \rho_{\text{DE}})} \sim \sqrt{\Lambda}.$$

This is the load-bearing structural success: the MOND scale is not a fitted input but a **horizon-derived** acceleration, locked to the dark-energy density. Three consequences follow immediately, all forced for a $w = -1$ (constant- Λ) cosmology:

- **a_0 is a cosmic constant at every epoch.** Because the bath clock is the dark-energy horizon and ρ_{DE} is flat, $a_0 = 9.36 \times 10^{-11}$ is identical at big-bang nucleosynthesis, recombination, cosmic noon, today, and in the infinite future. The MOND scale does not evolve (for $w = -1$).
- **The CMB acoustic epoch is deep-Newtonian.** A recombination-era fluid element has RMS acceleration $\sim 5 \times 10^{-8} \text{ m/s}^2 \gg a_0$, i.e. $x \sim 500 \gg 1$, so the modification is sub-percent and the acoustic peaks are untouched. The theory does not distort the CMB under constant a_0 .
- **The Machian limit is sharp.** As $\Lambda \rightarrow 0$, $cH_\Lambda \rightarrow 0$, $a_0 \rightarrow 0$, so $x \rightarrow \infty$ for every finite acceleration and $\mu_{\text{fw}} \rightarrow 1$ everywhere: pure Newton, no MOND. The horizon sources 100% of the low-acceleration inertia deficit, while the rest-mass coupling survives. The theory is, precisely and by its own statement, *half-Machian*.

What §2 does **not** do is fix Z , and therefore it does not fix a_0 's numerical value. The scale's *form* ($\sim \sqrt{\Lambda}$) is forced; its *normalization* is the one free parameter. This boundary is held throughout (§6, Quarantine).

3. The derived form: μ_{fw} and the convex ghost-free potential it minimizes

The relational excess of Input B is linear in a (§1.2). To turn that excess into the body's inertia ratio $\mu = m_{\text{inertial}}/m_{\text{rest}}$ one needs a constitutive shape — the theory's $\sqrt{\cdot}$ -equation of state,

$$g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0)} \text{ (the framework's own interpolation).}$$

Solving this for the inertia ratio $\mu = g_{\text{bar}}/a$ as a function of $x = a/a_0$ gives, with no further input (sympy-verified),

$$\mu_{\text{fw}}(x) = (\sqrt{1 + 4x^2} - 1) / (2x),$$

with the correct limits $\mu \rightarrow x$ deep ($a \ll a_0$, inertia strongly lowered) and $\mu \rightarrow 1$ Newtonian ($a \gg a_0$), approaching 1 strictly from below as $1 - \mu_{\text{fw}} \rightarrow a_0/2a$ with **no high-acceleration ceiling**. Written in the relational excess variable $u = 2Z\sqrt{(T^2 - T_0^2)}/T_0$ — which collapses *exactly* to $u = 2x$ because $T^2 - T_0^2 = (\hbar a/2\pi k_{\text{Bc}})^2$ and $T_0 = (\hbar/2\pi k_{\text{Bc}}) \cdot Z a_0$ — the same function is

$$m_{\text{I}}/m_{\text{rest}} = (\sqrt{1+u^2} - 1)/u = \tanh(\frac{1}{2} \cdot \text{asinh}(u)), u = 2x,$$

reproducing μ_{fw} to machine precision (residual 0.0 over seven decades in x ; the $\tanh$ - asinh identity verified to $\sim 10^{-38}$ in `mpmath`). The cleanest statement of the law is its inverse:

$$\mu / (1 - \mu^2) = a / a_0 = x \text{ (sympy-exact).}$$

Two structural facts make this a *constitutive theory*, not merely a suggestive curve:

- **It minimizes a convex, ghost-free potential.** μ_{fw} is the unique stable minimum of the Landau/Legendre potential $\Phi(\mu; x) = \mu^2/2 - x \cdot \mu + x \cdot \mu^3/3$, since $d\Phi/d\mu = 0 \iff \mu/(1-\mu^2) = x$ (the inverse identity above) and $\Phi'' = 1 + 2x\mu > 0$ everywhere. The equation of state is the extremum of a convex potential bounded below — the response is single-valued and stable.
- **It is causal and free of the Ostrogradsky ghost.** As an *algebraic* state law $\bar{g} = G(a)$ — a constitutive closure like $p = p(\rho)$ — the equation of motion is the ordinary second-order $\ddot{x} = \sqrt{(\bar{g}_{\text{bar}}^2 + \bar{g}_{\text{bar}} \cdot a_0)}$ with $\bar{g}_{\text{bar}} = GM/r^2$. There is no acceleration inside the law, hence no higher-derivative ghost; this is precisely how the theory evades the local-gating trap (gating inertia by $|a|$ in the action would put $\ddot{x}$ in the Lagrangian and produce a ghost; an equation of state does not). The force $\leftrightarrow$ acceleration map is monotone and invertible ($d\bar{g}_{\text{bar}}/dx = 2x/\sqrt{4x^2+1} > 0$), and the associated response energy is convex.

A useful exact fact at the scale itself: $\mu_{\text{fw}}(1) = (\sqrt{5} - 1)/2 = 1/\varphi = 0.61803\dots$, the inverse golden ratio. A body at $a = a_0$ retains exactly φ^{-1} of its rest inertia.

Honesty marker. The clean form lives in the *kinematic* excess variable $\sqrt{(T^2 - T_0^2)} \sim a$, not in the plain excess heat $\Delta T = T(a) - T_0$. The literal “ $\bar{g}_{\text{bar}} \propto \Delta T$ ” heat law matches deep-MOND ($\Delta T \propto a^2$ there, matching deep-MOND $\bar{g}_{\text{bar}} \propto a^2$) but then **overshoots to $\sim 11\times$** at high acceleration. So “inertia is a natural function of the heat” is true only in the deep limit; the full law is natural as a function of the kinematic excess (which is a in disguise). The $\sqrt{}$ -shape is a posit (Input A’s equation-of-state reading), not forced by the Machian principle, which fixes only the *argument* (excess-over-floor) and the boundaries $g(0) = 0, g(\infty) = 1$.

4. The derived kernel: $\theta(0) = \sqrt{2}$ and the Lorentzian shape

Because the theory is modified **inertia** and the bath is nonlocal in time, a body’s inertia depends on its recent acceleration *history*, not only the instantaneous value. In an external field this is encoded by a history kernel $\theta(y)$ entering the effective acceleration argument $A = a_{\text{internal}} +$

$\theta(y) \cdot a_{\text{external}}$, where $y = \omega_{\text{external}}/\omega_{\text{internal}}$ is the ratio of the external (orbital) frequency to the body's internal frequency. Two features of θ are fixed by the bath itself.

The DC weight $\theta(0) = \sqrt{2}$. Two independent framework-internal routes converge. First, the excess-heat engine: the inertia tracks the excess $\sqrt{(T^2 - T_0^2)}$, which is degree-1 (linear) in the acceleration amplitude, so θ multiplies an amplitude, not an energy; the -3 dB corner of an amplitude transfer sits at $1/\sqrt{2}$, giving $\theta(0) = \sqrt{2}$. (The alternative $\theta(0) = 2$ would require re-admitting the rejected T^2/energy reading, or a second independent memory clock — and the theory has a single Λ , a single dS clock.) Second, the additive-acceleration construction (Luo 2026): by the equivalence principle a static external field enters the first moment of the bath response at unit weight, while the variance-subtraction channel is owned by Λ alone — again the amplitude branch, again $\sqrt{2}$.

The Lorentzian shape. The de Sitter worldline correlator (Wightman function) is $W(u) = -(\kappa^2/16\pi^2)/\sinh^2(\kappa(u-i\epsilon)/2)$, $\kappa = 2\pi T_{\text{eff}}$. Its large-separation envelope is exponential, $e^{(-\kappa|u|)}$, which is precisely a Lorentzian transfer $1/(1 + (\omega/\omega_c)^2)$ in the frequency domain (with a y^{-2} tail). The kernel's *form* is therefore correlator-forced, no longer a bare ansatz.

So the kernel reads $\theta(y) = \theta_0/(1 + (\theta_0 - 1)y^2)$ with $\theta_0 = \sqrt{2}$ and $\theta(1) = 1$, decreasing monotonically with an exact zero-crossing of the *relational effect* at $y = 1$.

Honesty marker. What the de Sitter correlator does **not** hand over is the *location* of the corner. Taken literally the bath scale $\kappa \sim H_\Lambda$ is far below every orbital frequency, so the corner at $y = 1$ (external frequency = internal frequency) rides on the “internal orbit = the averaging bandwidth” postulate, licensed by Milgrom (1994) only in the quasi-static limit; the general multi-frequency case is obstructed, not closed. A memory-order residual also survives: a single-pole memory gives $(\theta(0) = \sqrt{2}, \text{tail } y^{-1})$ and a two-pole gives $(\theta(0) = 2, \text{tail } y^{-2})$; KMS alone does not fix the pole count. The honest classification is **plausible-with-forced-core**: existence of the corner, the DC value $\sqrt{2}$, and the Lorentzian form-class are bath-derived; the corner *location* and the memory order are modeling choices.

5. The sign — the relational principle selects MOND (conditional on Mach)

This is the deepest result and the honest crux of the theory, so it is stated carefully and two-sided.

There are two readings of the bath response, mirror images across $\mu = 1$:

- the **absolute** reading $R_{\text{abs}} = T/T_0$, which gives $R_{\text{abs}}(0) = 1$ — full inertial reaction assigned to an *unaccelerated* body. This is absolute-space inertia, and it is **identically** the passive-bath, dissipative-drag result one obtains from the Feynman–Vernon/Hu–Verdaguer influence functional built on the de Sitter KMS bath: that bath is passive (it satisfies detailed balance, sympy-verified), so its back-reaction kernel has Källén–Lehmann-positive spectral density and yields $\delta m \geq 0$ — inertia *raised* at low a , the **anti-MOND** sign.
- the **relational/excess** reading $R(u) = (\sqrt{1+u^2}-1)/u = \mu_{\text{fw}}$, which gives $R(0) = 0$ — zero dynamical inertia for an unaccelerated body. This is the **MOND** sign (inertia lowered at low a).

The relational/Machian principle of Input B does genuine selective work here, beyond mere naming. At $a = 0$ a body is geodesic — locally indistinguishable from rest in the cosmic frame

— so by the equivalence principle / Mach it feels **no inertial reaction**. The absolute reading $R_{\text{abs}}(0) = 1$ assigns full inertial reaction to that unaccelerated body, the very thing Mach and the equivalence principle deny. The Machian principle therefore supplies an **independent criterion** — not a redescription of “subtract the floor” — that rejects the absolute/anti-MOND reading and selects the excess/MOND reading. And the reading it kills is *exactly* the passive-bath influence-functional result that has otherwise been forcing anti-MOND. The relational excess response reproduces μ_{fw} to literally zero (sympy), in closed form $\tanh(\frac{1}{2} \operatorname{asinh}(2x))$.

This is why the passivity→anti-MOND theorem does not bind the theory: the relational reading is a *state function* (an equation of state), not a dissipative kernel; the no-go theorem applies to the kernel category, which is the absolute reading, and Mach is the principled reason to discard that category. So the MOND sign is upgraded from a bare definitional posit (“define inertia to vanish at the floor”) to a **consequence of a named physical premise**.

The cost, paid to the millimeter (honesty marker). The selection is **conditional on the Machian premise**. Three independent checks confirm that nothing weaker forces the sign:

1. **No thermodynamic extremum forces it.** Every bath state variable — $T(a)$, entropy $S(a)$, excess heat $Q(a)$ — is even in a ($T \sim \sqrt{a^2 + (cH\Lambda)^2}$). So any free-energy or entropy extremum yields $\mu \sim a^2$ at low a , the *wrong* deep-MOND slope ($\mu_{\text{fw}} \sim a$ is linear/odd). The linear drive that reproduces μ_{fw} is the floor-subtraction itself — the input, not the output. Stability *admits* the MOND sign (the anti-MOND potential is also locally convex) but does not *select* it.
2. **A two-bath non-equilibrium steady state breaks detailed balance but not passivity.** Adding a local field bath to the cosmic floor produces a nonzero steady heat current, yet the inertial shift remains a sum of two passive (positive) spectral densities $\Rightarrow \delta m \geq 0 = \text{anti-MOND}$ still. Breaking detailed balance redistributes energy; it never installs the negative spectral density (gain medium) the MOND sign would need.
3. **A temperature-independent sign theorem closes the active candidates.** For any KMS bath the detailed-balance factor $(1 - e^{(-\omega/T)})$ is strictly positive for all ω , $T > 0$, so $\operatorname{Im} \chi \geq 0$ independent of temperature — heating the bath cannot flip the dissipation sign. A growing de Sitter horizon is adiabatic versus any orbit by factors of 10^2 – 10^3 and, if it pumped at all, would have positive entropy production (dissipative = anti-MOND).

So the sign is derived **given Mach**, not by thermodynamics alone, and not by any passive mechanism. What was traded for the bare sign-posit is the Machian premise itself (that inertia is relational, and that the de Sitter vacuum frame is the cosmic rest frame). That is a genuine upgrade — a posit replaced by a deeper, more falsifiable premise carrying new content (a real universal floor temperature $T_0 = 2.20 \times 10^{-30}$ K) — but it is **not a free lunch**, and the theory states the sign as conditional on Mach.

The premise is not free-floating: it factors, and most of it grounds. The Machian premise decomposes into the *existence* of a physical relational reference and the *definition* of inertia as the excess over its floor. The first clause is **not a new posit** — the reference is the same de Sitter-vacuum / cosmic rest frame the theory already carries and is *already constrained on* at Cassini through the $s^{\wedge}\text{TX}$ dipole, and T_0 is the Gibbons–Hawking temperature of exactly that frame; the

equivalence-principle boundary (geodesics unmodified) then falls out for free, since the bath is flat at the floor, $dT/da|_0 = 0$, for *any* version of the law. What does **not** reduce is a single clause — the floor-subtraction, i.e. common-mode rejection of T_0 — and it is not derivable from renormalization either, for a precise reason. Renormalization mandates subtracting the $a = 0$ reference only for **gauge** quantities (energy, work), which enter dynamics through *differences*, so the shared constant T_0 cancels and the subtraction is forced and content-free. Inertial mass is **non-gauge**: it enters as $F = m a$, so its absolute level at $a = 0$ is physical (the absolute and relational readings differ at every $a > 0$ by exactly 1). Tellingly, the theory subtracts T_0 for inertia *alone* while retaining it for s^{TX} , for velocity relative to the frame, and as a *reported* floor temperature — exactly what a universal vacuum-subtraction principle would forbid. The foundation, stated precisely, is therefore: **the equivalence principle + the de Sitter bath + the (Cassini-constrained) preferred frame + one irreducible, falsifiable Machian premise** — *inertia is the work-doing excess over the cosmic floor; a body at rest in the vacuum carries no dynamical inertia* — respectable in the way the equivalence principle itself was once a stated premise, and with a clear reason (the non-gauge character of inertia) why it does not reduce further.

6. The one free parameter: Z

The theory has exactly one free number, the geometric normalization

$$Z = 2\sqrt{(8\pi/3)} = \sqrt{(32\pi/3)} = 5.7888, \text{ so that } a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2.$$

Z is **provably free** by the κ -closure result (DOI 10.5281/zenodo.20963226): the normalization κ of the relevant action is the overall scaling of a ϕ -linear term, and no-ghost/stability/sound-speed conditions, unitarity (signs/causal structure/ratios only), and holography (the sole dimensionless group cancels classically) are all invariant under it. Nothing among those structural requirements fixes Z . An attempt to fix it microscopically (the CKN ultraviolet–infrared bound, matched to a Standard-Model degree-of-freedom count) returns *not-fixed*: the honest count gives an $O(1)$ coefficient that is a function of the degree-of-freedom number, whereas the theory’s normalization is pure geometry (Friedmann-3, Einstein- 8π), the no-content (degree-of-freedom = 1) limit. Z is geometric, not microscopic, and it is the allowed single free number of a one-parameter theory.

The right analogy is Newton’s constant in general relativity. GR derives the *structure* of gravity — the field equations, the geodesic law, light bending, the perihelion shift — with one dimensionful constant G that the theory does not predict and that must be measured. Here the theory derives the *structure* of the MOND scale — its $\sim \sqrt{\Lambda}$ form, the interpolation, the kernel, the sign — with one normalization Z that the theory does not predict. **a_0 ’s numerical value is not derived; its form and all its consequences are.** This is what makes the proposal a one-parameter theory rather than a zero-parameter one, and it is stated plainly rather than hidden.

7. Predictions

The theory’s live content is empirical. Four predictions, ordered roughly by decisiveness:

1. **An s^{TX} Lorentz-violation dipole.** A preferred-frame modified-inertia theory is, in the Standard-Model-Extension language, a gravity-sector background $s_{\mu\nu}$. The same $a_0/|a|$ that produces MOND induces a computable boost-dipole coefficient s^{TX} . The component-by-component value sits at roughly Saturn 8.68×10^{-10} , about $1.5\times$ the tightest current planetary-ephemeris (INPOP/Cassini) bound — the more decisive near-term gravity-side test, analysis-limited over $\sim 2028\text{--}2032$. The companion CPT-even-only theorem (Lorentz-violating but CPT-preserving) is a structural prediction of the same construction.
2. **The $a_0(z)$ hostage.** Because $a_0 \propto \sqrt{(\rho_{\text{DE}}(z))}$, a_0 is exactly constant if $w = -1$ but evolves if the dark-energy equation of state does. Under the DESI/CPL fit ($w_0 = -0.752$, $w_a = -0.86$) the scale shows a phantom-divide bump of $\sim +6\%$ near $z = 0.405$ and a steep decline at high z ($\sim -26\%$ by $z = 3$), testable as a Baryonic-Tully-Fisher-zero-point / rotation-curve normalization drift. This prediction **dissolves entirely if $w \rightarrow -1$** — it is a hostage to the equation of state, stated as non-diagnostic until the data sharpen (DESI DR3 gate 2026–27; ELT tracking early-mid 2030s).
3. **A modified-gravity-impossible dwarf orbital-history dispersion.** Because inertia here is a functional of acceleration *history*, a diffuse dwarf galaxy on a radial-plunge orbit should run hotter (larger internal σ) than a circular-orbit dwarf of the same mass and same closest approach. At the selected $\theta(0) = \sqrt{2}$ the boost is $+12\text{--}14\%$ in the realistic plunging band (a $+12\text{--}30\%$ envelope spanning the memory-order residual), with an exact zero-crossing at $y = 1$. This is **modified-gravity-impossible**: in any metric or field MOND (AQUAL/QUMOND/AeST) and in Λ CDM the external-field effect is instantaneous (internal dynamics depend only on the *momentary* external acceleration), so those theories predict exactly zero σ -vs-history correlation at fixed pericenter, for any a_0 . The named carriers reaching the non-adiabatic band are Crater II ($y = 3.28$) and Antlia II ($y = 2.55$) against adiabatic controls Fornax and Sculptor. The decisive test is Gaia DR4 (December 2026) plus diffuse-carrier spectroscopy. A pre-registered pilot on 24 Milky-Way dwarfs is a statistically underpowered **null** (partial Spearman $\rho = -0.196$, $p = 0.395$) — existing data cannot yet test it (DOI 10.5281/zenodo.20947913).
4. **The floor temperature $T_0 = 2.20 \times 10^{-30}$ K.** The theory predicts a real, universal de Sitter floor temperature shared by every body — the Gibbons–Hawking temperature of the cosmic horizon — as the athermal reference from which dynamical inertia is measured. It is the new falsifiable content the Machian upgrade carries.

8. Honest scope and open doors

The two-sided ledger, repeated and explicit, because the theory’s credibility rests on it:

- **The sign rests on the equivalence principle, the bath, the Cassini-constrained preferred frame, and one irreducible Machian premise.** §5 shows the premise *factors*: its relational *reference* is the framework’s own (already s^{TX} -constrained) preferred frame, and the equivalence-principle boundary is automatic from the bath’s flat floor — neither is a new posit. The single clause that does not reduce is the *floor-subtraction* (common-mode rejection of T_0), and it is provably not forced by renormalization (which mandates subtracting the $a = 0$ reference only for **gauge** quantities; inertial mass is **non-gauge**) nor by any thermodynamic

extremum or non-equilibrium steady state. This is the honest crux: a single, named, falsifiable premise — not a free lunch, and now with a clear reason (inertia’s non-gauge character) why it is irreducible.

- **Z, and hence a_0 ’s value, is not derived.** Z is provably free (κ -closure). The theory is one-parameter, not zero-parameter; a_0 ’s *form* $\sim \sqrt{\Lambda}$ is forced, its *normalization* is the one measured number.
- **The Standard Model is walled — this is NOT a TOE.** The kernel fixes one infrared scale ($a_0 \sim cH_\Lambda$). It does not count, quantize, or pattern Standard-Model masses; the flavor-disjoint-representation and forced-kernel walls stand. The author has publicly retracted (2026) all earlier TOE / Standard-Model-derivation overclaims; the honest position is the a_0 reframing only, and this paper makes no claim beyond it.

The named open doors, stated so that none is dressed as closed:

1. **Ground the floor-subtraction.** The relational *reference* and the equivalence-principle boundary now reduce to the framework’s Cassini-constrained preferred frame and the bath; the one clause still standing is *why* inertia is the **work-doing excess** over the cosmic floor — i.e. why common-mode rejection of T_0 beats the absolute/passive-bath reading. This is shown *not* forced by renormalization (gauge vs non-gauge) or by thermodynamics, so a first-principles modified-inertia reason for it is the genuine remaining theory door.
2. **Establish the $\sqrt{}$ -shape’s necessity.** The interpolation form is the equation-of-state posit; Mach fixes only the argument and the boundary conditions, not the $\sqrt{}$ -shape.
3. **The named active pump.** The lone un-theorem’d reopener on the kernel side is an in-band active galactic mechanism that would break de Sitter passivity in-band — which would also have to explain a_0 ’s observed universality. Not supplied by any passive source examined here.
4. **The kernel corner location and memory order** (§4) remain quasi-static modeling choices.

The theory is consistent with, and is the capstone of, the previously published effective-field-theory results: the de Sitter–Unruh a_0 no-go and one-free-number papers (DOIs 10.5281/zenodo.20935948, 20938891), the κ -closure / one-free-number result (DOI 10.5281/zenodo.20963226), and the dwarf orbital-history prediction (DOI 10.5281/zenodo.20947913). It claims to be exactly what those papers, and the retraction, support: a one-parameter modified-inertia account of a single infrared scale — **a theory, in the one-parameter sense, not a theory of everything.**

Reproducibility note

All symbolic and numerical results above were verified with sympy, numpy, and mpmath: the linear excess identity $\sqrt{(T^2 - T_0^2)} = (\hbar/2\pi k_B c) \cdot a$ and the floor universality $dT_0/da = 0$ (Input B); the equation-of-state solve $\mu_{fw} = (\sqrt{(1+4x^2)}-1)/2x$ with the inverse $\mu/(1-\mu^2) = x$, the variable collapse $u = 2x$, and the identity $(\sqrt{(1+u^2)}-1)/u = \tanh(\frac{1}{2} \operatorname{asinh} u)$ to $\sim 10^{-38}$ (§3); the convex potential $\Phi = \mu^2/2 - x\mu + x\mu^3/3$ with unique stable minimum at μ_{fw} and $\Phi'' = 1 + 2x\mu > 0$, and the monotone-invertible, ghost-free force map (§3); the $\theta(0) = \sqrt{2}$ amplitude-branch selection and the Lorentzian/exponential-memory envelope of the de Sitter $1/\sinh^2$ correlator (§4); the KMS detailed-balance relation $S(-\omega)/S(\omega) = e^{(-\beta\omega)}$, the passive $\delta m \geq 0$ result

both spectrally and on the $1/\sinh^2$ kernel, the two-bath steady-current-but-still-passive result, and the temperature-independent $(1 - e^{(-\omega/T)}) > 0$ sign theorem (§5); and the κ -closure invariance establishing Z as free (§6). Footing throughout: $a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2$, $Z = \sqrt{(32\pi/3)} = 5.7888$, $cH_\Lambda = 5.42 \times 10^{-10} \text{ m/s}^2$, $T_0 = 2.20 \times 10^{-30} \text{ K}$, and the framework's own interpolation μ_{fw} and Deser–Levin $T(a) - \text{never McGaugh's } v$. Supporting scripts accompany the source memoranda (DERIVE_THE_SIGN, CONSTITUTIVE_LAW_SWING, INFLUENCE_FUNCTIONAL_DELTAT_INERTIA, THETA_KERNEL_TOWARD_FORCED, MAXIMAL_EXTRAPOLATION).

Framework-internal throughout; no rival-theory comparison is used to judge the theory. Both-sided honesty (forced vs posited) is stated at every step. Not a theory of everything.